Norms and Metrics

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1 Introduction

A vector space gives us addition and scalar multiplication, but no notion of *length* or *distance*. The moment we want to talk about convergence of an iterative algorithm, continuity of a loss function, or generalisation bounds on a hypothesis class, we need to measure how close two vectors are. *Norms* and *metrics* provide exactly this quantitative geometry.

The two notions are tightly related: every norm on a vector space induces a metric in a canonical way. The converse fails, and the failure is instructive — it shows that metrics are strictly more general than norms, and that some legitimate measures of distance (e.g. the discrete metric, or the Kullback–Leibler divergence after suitable repair) cannot be derived from any notion of vector length.

2 Definitions

Throughout, let $\mathbb{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let V be a vector space over $\mathbb{F}$.

Definition 1 (Norm)**.** A function $\|\cdot\|\colon V \to \mathbb{R}$ is a *norm* if for every $x, y \in V$ and every scalar $\alpha \in \mathbb{F}$,

1. (Positive definiteness) $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$.
2. (Absolute homogeneity) $\|\alpha x\| = |\alpha|\,\|x\|$.
3. (Triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$.

The pair $(V, \|\cdot\|)$ is called a *normed vector space*.

Definition 2 (Metric)**.** Let X be a set. A function $d\colon X \times X \to \mathbb{R}$ is a *metric* (or *distance function*) if for every $x, y, z \in X$,

1. (Non-negativity) $d(x, y) \geq 0$.

2. (Identity of indiscernibles) $d(x, y) = 0 \iff x = y$.
3. (Symmetry) $d(x, y) = d(y, x)$.
4. (Triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is called a *metric space.*

Definition 3 (Induced metric)**.** Let $(V, \|\cdot\|)$ be a normed vector space. The *metric induced by* $\|\cdot\|$ is

$$d(x, y) := \|x - y\|, \qquad x, y \in V.$$

Example 1 (ℓ^p norms on $\mathbb{R}^n$)**.** For $1 \leq p < \infty$ and $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, define

$$\|x\|_p := \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}, \qquad \|x\|_\infty := \max_{1 \leq i \leq n} |x_i|.$$

Each $\|\cdot\|_p$ is a norm. The triangle inequality for $p \geq 1$ is the *Minkowski inequality.*

3 Main theorem

Theorem 1 (Norms induce metrics)**.** *Let $(V, \|\cdot\|)$ be a normed vector space. Then the function*

$$d \colon V \times V \to \mathbb{R}, \qquad d(x, y) := \|x - y\|,$$

is a metric on V.

Proof. We verify the four axioms of a metric.

(1) Non-negativity. By positive definiteness of the norm, $d(x, y) = \|x - y\| \geq 0$ for all $x, y \in V$.

(2) Identity of indiscernibles. We use positive definiteness again:

$$d(x, y) = 0 \iff \|x - y\| = 0 \iff x - y = 0 \iff x = y.$$

(3) Symmetry. By absolute homogeneity with $\alpha = -1$,

$$d(x, y) = \|x - y\| = \big\|(-1)\,(y - x)\big\| = |-1|\,\|y - x\| = \|y - x\| = d(y, x).$$

(4) Triangle inequality. For any $x, y, z \in V$, write $x - z = (x - y) + (y - z)$. The triangle inequality for the norm gives

$$d(x, z) = \|x - z\| = \big\|(x - y) + (y - z)\big\| \leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z).$$

All four axioms hold, so d is a metric on V. $\square$

4 A metric that is not norm-induced

Proposition 2 (Discrete metric)**.** *Let V be a nonzero vector space and define*

$$d(x,y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

Then d is a metric on V, but there is no norm $\|\cdot\|$ on V such that $d(x,y) = \|x-y\|$ for all x, y.

Proof. That d is a metric is a direct check of the four axioms.

Suppose for contradiction that $d(x,y) = \|x-y\|$ for some norm $\|\cdot\|$. Pick any $x \neq 0$. By homogeneity,

$$\|2x\| = 2\,\|x\|.$$

On the other hand, $2x \neq 0$ and $x \neq 0$, so

$$\|2x\| = d(2x, 0) = 1 \qquad \text{and} \qquad \|x\| = d(x,0) = 1.$$

Combining the displays gives $1 = 2$, a contradiction. Hence d does not come from any norm. $\square$

5 Why this matters

Once a metric is fixed, the entire machinery of analysis is unlocked: open and closed sets, continuity, convergence, Cauchy sequences, completeness, compactness, Lipschitz constants. In machine learning, the choice of norm is rarely innocent: ℓ^1 regularisation produces sparse solutions, ℓ^2 regularisation shrinks coefficients isotropically, and the spectral (operator) norm controls how much a neural-network layer can amplify inputs. The Wasserstein distance is a genuine metric on probability distributions and underlies optimal transport; the Kullback–Leibler divergence is *not* symmetric and *not* a metric, even though it is often informally called a "distance".